

Question number	Answer	Additional guidance	Mark										
8(a)	An answer that includes the following points: <table><tr><th>Part of Brain</th><th>Function</th></tr><tr><td>cerebellum</td><td>co-ordination of (voluntary) movement</td></tr><tr><td>cerebral hemispheres</td><td>interpretation of information from retina</td></tr><tr><td>medulla oblongata</td><td>regulation of (cardiovascular / respiratory system)</td></tr><tr><td>(posterior) Pituitary</td><td>secretion of ADH</td></tr></table>	Part of Brain	Function	cerebellum	co-ordination of (voluntary) movement	cerebral hemispheres	interpretation of information from retina	medulla oblongata	regulation of (cardiovascular / respiratory system)	(posterior) Pituitary	secretion of ADH	Accept posture, balance, muscular activity accept if answers are descriptive accept cerebrum / visual cortex / occipital lobe/ cerebral cortex Accept hypothalamus Two correct cells = 1 mark All four cells = 2 marks	2
Part of Brain	Function												
cerebellum	co-ordination of (voluntary) movement												
cerebral hemispheres	interpretation of information from retina												
medulla oblongata	regulation of (cardiovascular / respiratory system)												
(posterior) Pituitary	secretion of ADH												

Question number	Answer	Additional guidance	Mark
8(b)	An answer that includes four of the following points: Similarity • both are (neuro-imaging) methods giving high resolution (1) • both give images of areas with plaques in the brain (1) Differences • fMRI expensive CT cheaper (1) • fMRI will show {metabolism / oxygen uptake / brain activity} CT only shows structures (1) • fMRI uses magnetic resonance CT uses X rays (1)	2 marks max for similarities, 2 marks for differences needs to be compare and contrast. ACCEPT fMRI 45 mins CT 5-10 mins fMRI does not use ionising radiation CT does (so increased risk of cancer developing) ignore fMRI has higher resolution than CT	4

Question number	Answer	Additional guidance	Mark
8(c)	An answer that includes four of the following points: • amyloid protein will cause bacteria to stick together(1) • enhancing phagocytosis of the bacteria / stimulates activity of white blood cells (1) • acts as an antibiotic / anti-microbial (1) • amyloid protein could disrupt ability of bacteria to adhere directly to host cell (1) • prevents entry of bacteria into cells (1)	ACCEPT agglutinate / sticks to the surface of the bacteria ACCEPT description of phagocytosis do not accept trigger immune response.	4

Question number	Answer	Additional guidance	Mark
8(d)	An answer that includes the following points: • bacteria carried in blood from (infected) gums (1) • bacteria penetrate the blood-brain barrier (1) • bacteria /bacterial products cause the release of {histamine/ chemicals / cytokines} (1) • resulting in vasodilation (1) • consequence of vasodilation (1)	ACCEPT bacteria can invade white blood cells which cross the blood brain barrier eg increased blood flow / oedema /increase in wbc in brain/increased oxygen.	4

Question number	Answer	Additional guidance	Mark
8(e)	An answer that includes three of the following points: • gingipains are enzymes (that break down proteins in membranes) (1) • degrading/break down cell membranes (1) • resulting in loss of (active) brain tissue (in the cerebrum) (1) • leads to reduction in memory (especially short-term memory) (1)	ACCEPT reference to an effect on transport channels within membrane ACCEPT in context effect on nerve impulses / action potentials	3

Question number	Answer	Additional guidance	Mark
8(f)	A description that includes three of the following points: • extraction of infected part of cerebral cortex (1) • use microarray (to identify mRNAs in sample) (1) • (analysis of mRNA) to identify genetic material from Porphyromonas (1) • use of bioinformatics / algorithms to analyse the (enormous quantity of) data (from the microarray) (1)	ACCEPT use of specific microarray eg oligo- nucleotide microarray	3

